

Exp 5 - Design of High Pass Filter using Windowing Techniques

5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

PROGRAM

```
clc; clear; close all;

%% --- Input Parameters ---

fp = 1000;    % High-pass cutoff frequency (Hz)
fs = 8000;    % Sampling frequency (Hz)
N = 50;      % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---

figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

OUTPUT

```
--- High-Pass Coefficients (Hanning Window) ---
-0.0000
  0
 0.0003
 0.0008
 0.0009
  0
-0.0020
-0.0038
-0.0035
  0
 0.0057
 0.0100
 0.0087
```

0
-0.0127
-0.0216
-0.0183
0
0.0267
0.0464
0.0410
0
-0.0726
-0.1568
-0.2243
0.7500
-0.2243
-0.1568
-0.0726
0
0.0410
0.0464
0.0267
0
-0.0183
-0.0216
-0.0127
0
0.0087
0.0100
0.0057
0
-0.0035
-0.0038
-0.0020
0
0.0009
0.0008

0.0003
0
-0.0000

